

# Industrial Internship Report on " Prediction of Agriculture Crop Production in India"

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## *Executive Summary*

This report presents the work completed during the Industrial Internship organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT).

During this internship, I worked on the project "**Prediction of Agriculture Crop Production in India**", which focuses on applying Machine Learning techniques to predict agricultural crop yield using cultivation and production-related factors. The project involved data collection, preprocessing, exploratory data analysis, feature engineering, model development, and performance evaluation.

A Random Forest Regression model was developed using agricultural data containing crop type, state information, cultivation costs, and production costs. The model achieved an **R<sup>2</sup> score of 0.9463**, demonstrating strong predictive performance.

This internship provided valuable exposure to Data Science and Machine Learning concepts, including data preprocessing, model training, evaluation metrics, and data visualization. It also enhanced my problem-solving skills and practical understanding of applying machine learning techniques to real-world agricultural challenges.

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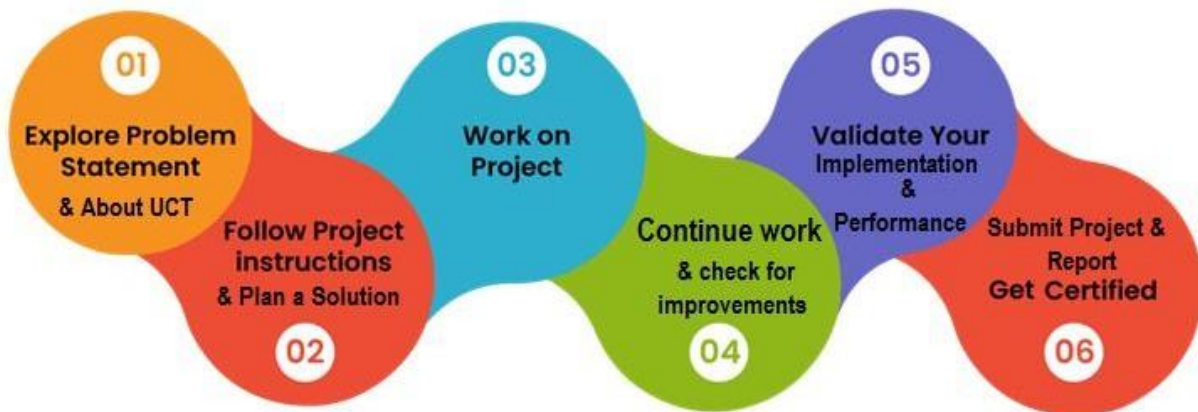
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## 1 Preface

This internship was conducted under **upskill Campus** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt Ltd (UCT)** to provide practical exposure to Data Science and Machine Learning concepts.

During the six-week internship, I worked on the project "**Prediction of Agriculture Crop Production in India**". The project involved analyzing agricultural data, preprocessing datasets, developing a Machine Learning model, and evaluating its performance for crop yield prediction.

This internship provided me with valuable experience in Python programming, data analysis, machine learning model development, and data visualization. It helped me understand how technology can be applied to solve real-world problems in the agriculture sector.



### My Learnings and Overall Experience

I gained hands-on experience in data preprocessing, feature engineering, model training, and performance evaluation. The internship enhanced my analytical thinking and problem-solving skills.

### Acknowledgement

I would like to thank **UCT**, **upskill Campus**, **The IoT Academy**, and **KIIT University** for providing this learning opportunity and supporting me throughout the internship.

### Message to Juniors and Peers

I encourage students to participate in internships and project-based learning programs as they provide practical knowledge and industry exposure beyond classroom learning.

## 2 Introduction

### 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



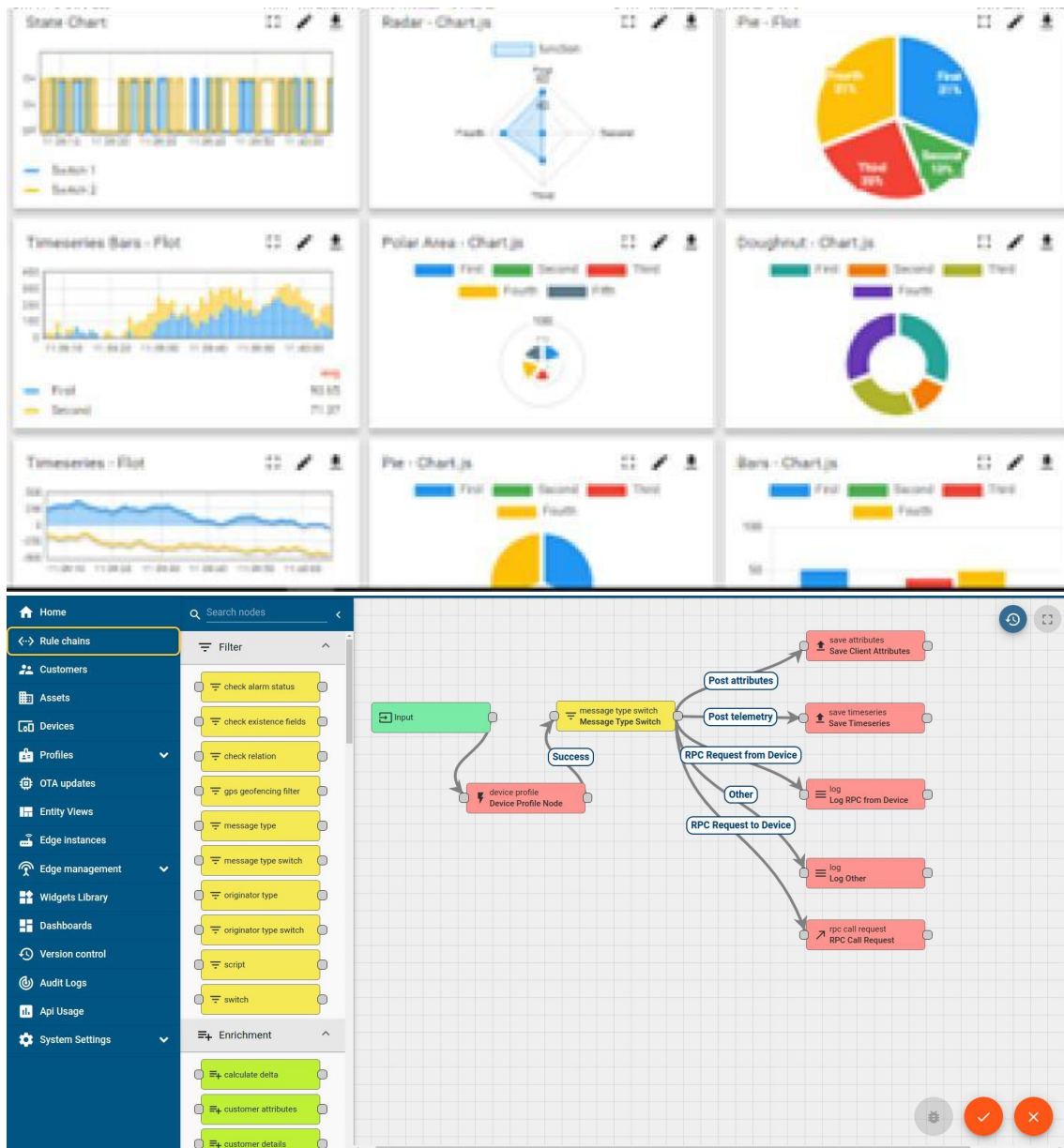
#### i. UCT IoT Platform ( )

**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





## FACTORY WATCH

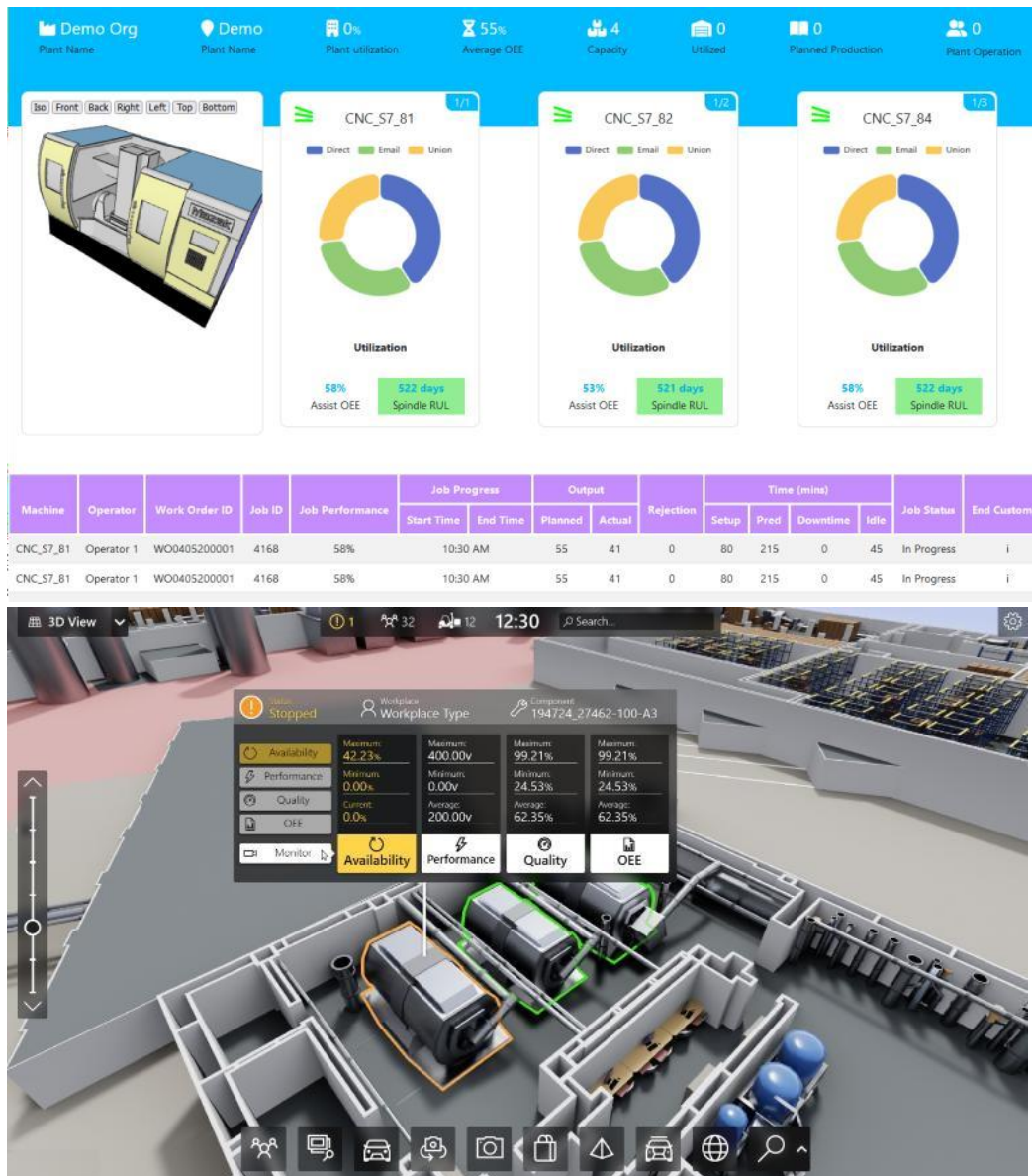
### ii. Smart Factory Platform ( )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



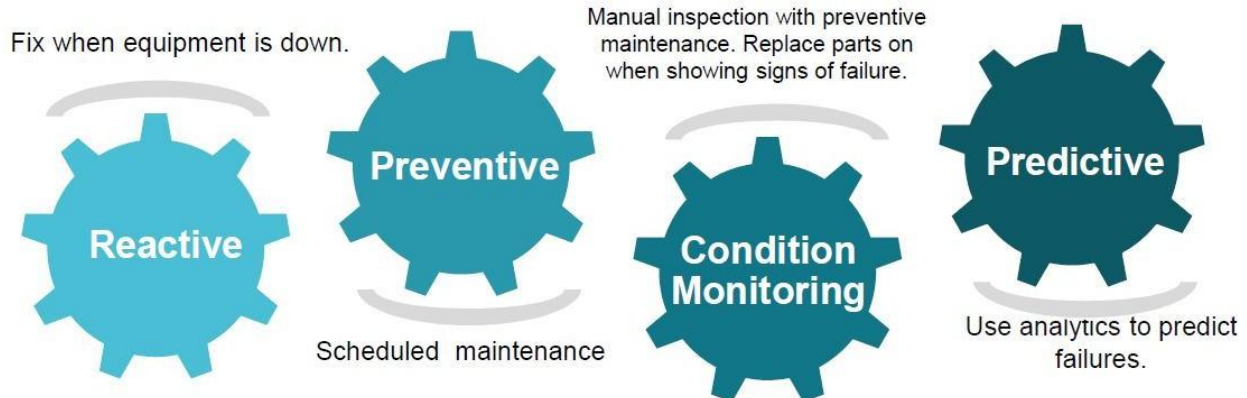


### iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

### iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

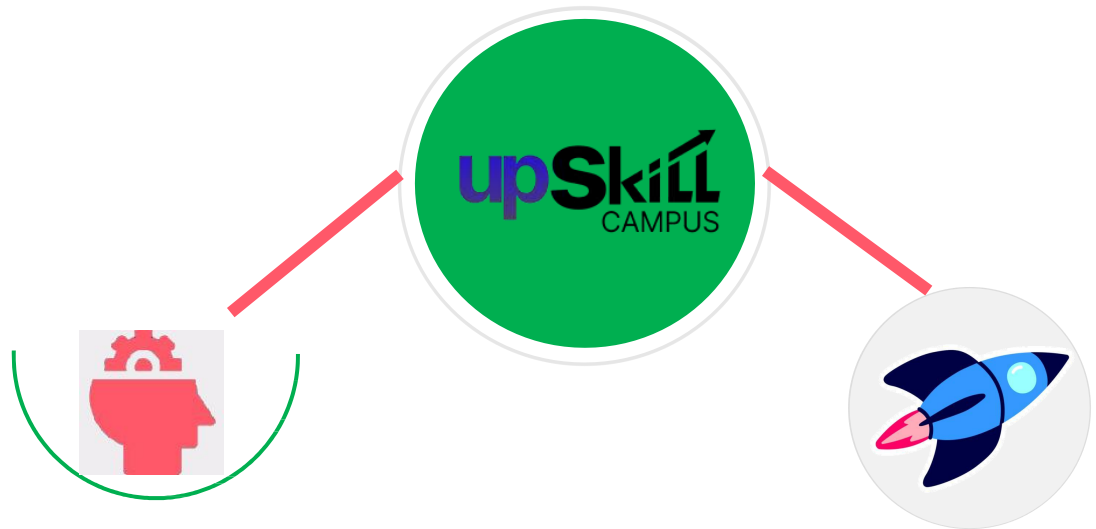


## 2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

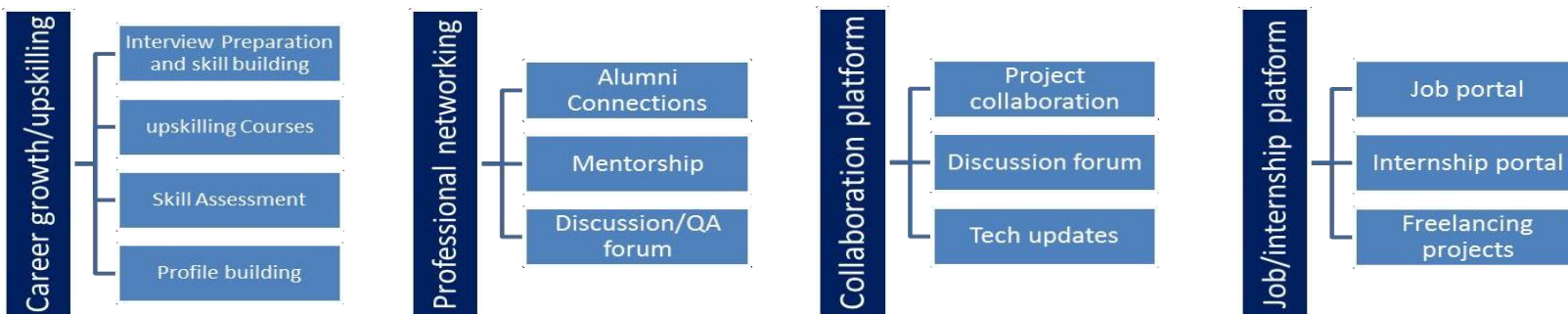




Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



## 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

## 2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

## 2.5 Reference

[1] Scikit-Learn Documentation – <https://scikit-learn.org>

[2] Pandas Documentation – <https://pandas.pydata.org>

[3] Matplotlib Documentation – <https://matplotlib.org>

## 2.6 Glossary

| Terms                        | Acronym        |
|------------------------------|----------------|
| Machine Learning             | ML             |
| Random Forest                | RF             |
| Mean Absolute Error          | MAE            |
| Root Mean Square Error       | RMSE           |
| Exploratory Data Analysis    | EDA            |
| Coefficient of Determination | R <sup>2</sup> |

### 3 Problem Statement

Agriculture is one of the most important sectors of the Indian economy and plays a vital role in food production and employment. Accurate prediction of crop production can help farmers, agricultural organizations, and policymakers make informed decisions regarding cultivation planning, resource allocation, and productivity improvement.

The objective of this project is to develop a Machine Learning model capable of predicting agricultural crop production based on various factors such as crop type, state, cultivation cost, and production cost. The system analyzes historical agricultural data and identifies patterns that influence crop yield.

By applying data preprocessing, feature engineering, and machine learning techniques, the proposed solution aims to provide reliable crop yield predictions that can support better agricultural planning and decision-making

## 4 Existing and Proposed solution

### Existing Solution

Traditional methods for estimating crop production are generally based on manual surveys, historical records, and expert analysis. These approaches are often time-consuming, require significant human effort, and may not accurately capture the complex relationships between cultivation costs, crop types, and yield.

### Proposed Solution

The proposed solution uses a Machine Learning-based approach to predict agricultural crop production. Historical agricultural data containing crop type, state information, cultivation costs, and production costs is analyzed using a Random Forest Regression model. The model learns patterns from the data and predicts crop yield with high accuracy.

### Value Addition

- Faster and automated prediction process.
- Data-driven agricultural decision making.
- Improved prediction accuracy.
- Better resource planning and crop management.
- Can be extended with weather, soil, and rainfall data in the future.

### 4.1 Code submission (Github link):

Code Link: [Link](#)

### 4.2 Report submission (Github link) :

Report Link: [Link](#)

## 5 Proposed Design/ Model

The proposed system follows a Machine Learning workflow for predicting agricultural crop production. The agricultural dataset is first preprocessed and transformed into a suitable format. Feature encoding and data preparation techniques are applied before training the Random Forest Regression model. The trained model is then evaluated using performance metrics and used for crop yield prediction.

### 5.1 High Level Diagram (if applicable)

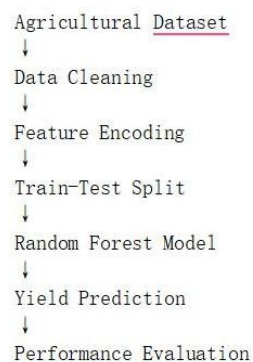


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

### 5.2 Low Level Diagram (if applicable)

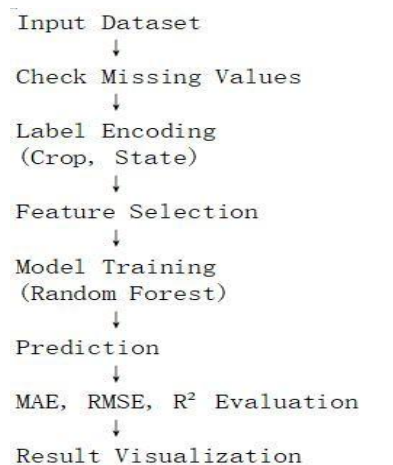


Figure 2: Low Level Diagram of System Workflow



### 5.3 Interfaces (if applicable)

- **Dataset Interface**

Input dataset containing crop information, state details, cultivation cost, production cost, and yield.

- **Processing Interface**

Data preprocessing using Pandas.

Feature encoding using Label Encoder.

Model training using Random Forest Regressor.

- **Output Interface**

Predicted crop yield values.

Performance metrics (MAE, RMSE,  $R^2$ ).

Visualization graphs:

- Yield by Crop
- Yield by State
- Actual vs Predicted Yield

## 6 Performance Test

The developed Machine Learning model was evaluated using standard regression metrics to measure prediction accuracy and model effectiveness. The main objective was to predict agricultural crop yield based on crop type, state information, and cultivation-related costs.

The primary constraints considered during model development were:

- Small dataset size (49 records).
- Variability in crop yield across different states and crops.
- Limited number of input features.
- Requirement for accurate prediction with minimal computational cost.

To address these constraints, data preprocessing, feature encoding, and Random Forest Regression were used to improve prediction performance

### 6.1 Test Plan/ Test Cases

| Test Case          | Description                   | Expected Result                  |
|--------------------|-------------------------------|----------------------------------|
| Data Loading       | Load agricultural dataset     | Dataset loaded successfully      |
| Data Preprocessing | Handle categorical features   | Features encoded correctly       |
| Model Training     | Train Random Forest model     | Model trained successfully       |
| Prediction         | Generate yield predictions    | Predicted yield values generated |
| Evaluation         | Calculate performance metrics | MAE, RMSE, and $R^2$ obtained    |

### 6.2 Test Procedure

- Load and inspect the agricultural dataset.
- Remove inconsistencies and preprocess data.
- Apply Label Encoding on Crop and State attributes.
- Split data into training and testing sets.

- Train the Random Forest Regression model.
- Generate predictions on test data.
- Evaluate model performance using MAE, RMSE, and  $R^2$  metrics.
- Visualize the results using graphs

### 6.3 Performance Outcome

**Metric Value**

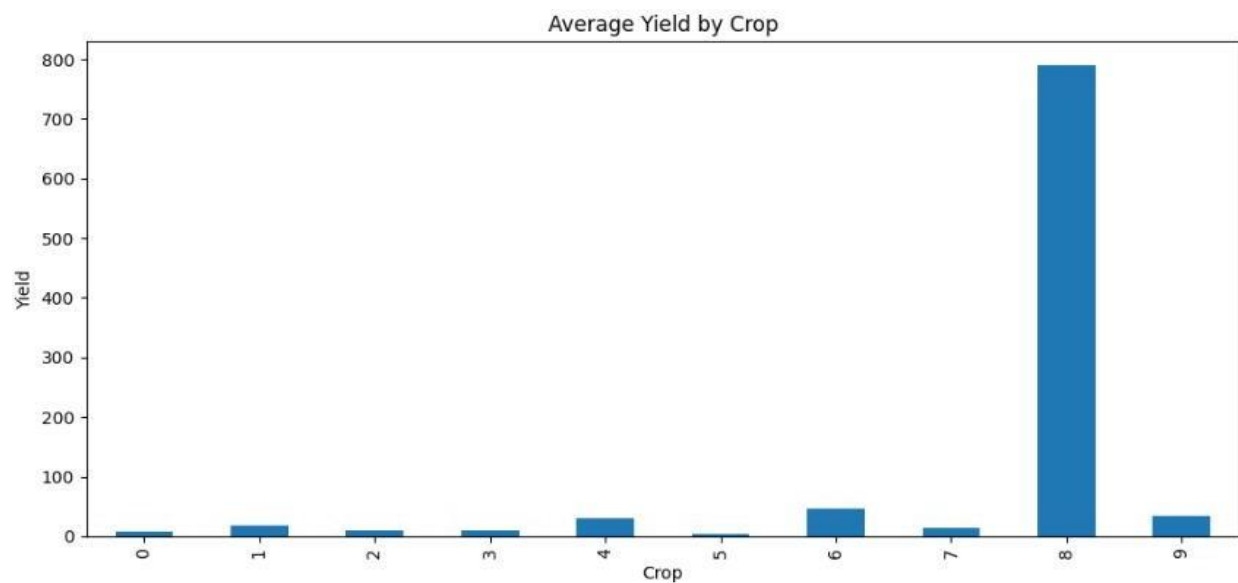
MAE 28.44

RMSE 69.66

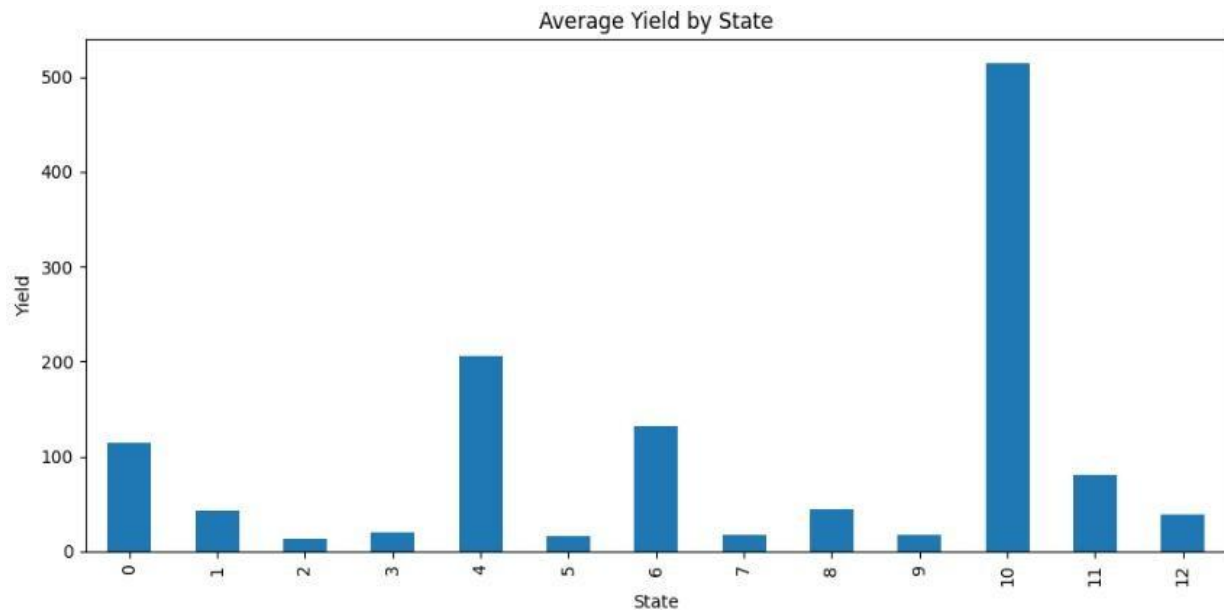
$R^2$  Score 0.9463

#### Result Analysis

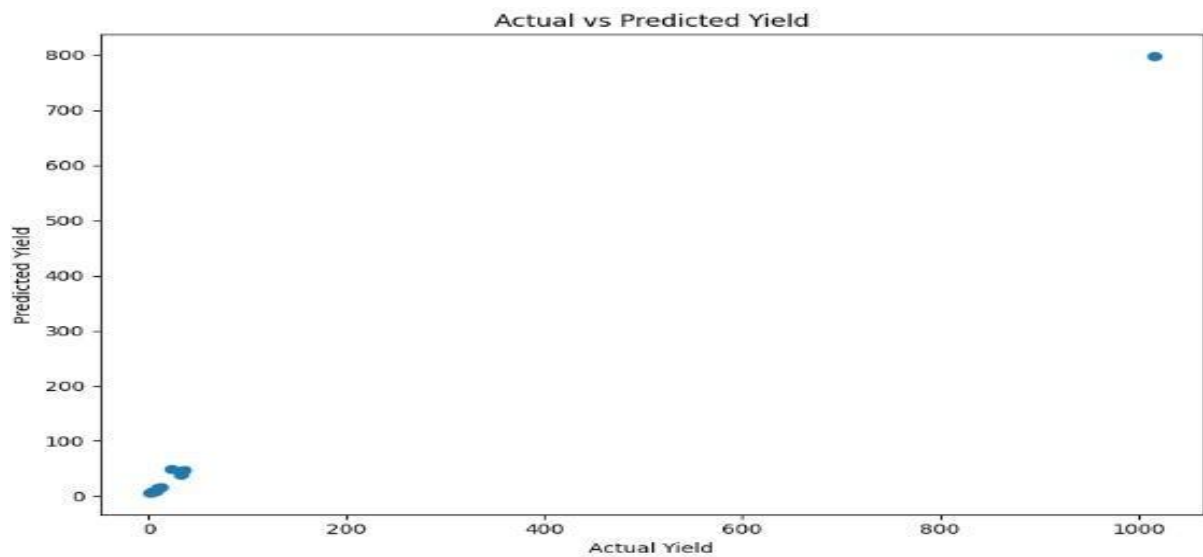
The model achieved an  **$R^2$  score of 0.9463**, indicating that approximately **94.63% of the variance in crop yield** was explained by the model. The obtained MAE and RMSE values demonstrate good predictive performance considering the limited dataset size



**Figure 3: Average Yield by Crop**



**Figure 4: Average Yield by State**



**Figure 5: Actual vs Predicted Yield**

## 7 My learnings

During this internship, I gained practical exposure to the complete Machine Learning project lifecycle, including data collection, preprocessing, feature engineering, model development, evaluation, and result visualization.

I improved my skills in:

- Python programming for data analysis.
- Data preprocessing using Pandas.
- Data visualization using Matplotlib.
- Machine Learning model development using Scikit-Learn.
- Feature encoding and model evaluation techniques.
- Applying Machine Learning to solve real-world agricultural problems.

Working on the project "**Prediction of Agriculture Crop Production in India**" helped me understand how data-driven solutions can be used to support decision-making in the agriculture sector. It also strengthened my analytical thinking, problem-solving abilities, and project development skills.

This internship has increased my confidence in building end-to-end Machine Learning projects and has provided valuable experience that will contribute to my future academic and professional growth.

## 8 Future work scope

- Integrate weather and rainfall information to improve prediction accuracy.
- Include soil quality and fertilizer usage data as additional features.
- Train the model on larger agricultural datasets covering more regions and years.
- Compare advanced Machine Learning algorithms such as XGBoost and Gradient Boosting.
- Develop a web-based dashboard for real-time crop production prediction and visualization.
- Extend the system to provide crop recommendations based on environmental and economic factors.